



## AQ1.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

### 4.1 Level 1:

Let  $A$  be a  $3 \times 3$  matrix with non-zero determinant. If  $\det(2A) = k \det(A)$ , then what will be the value of  $k$ ?

[Hint: If a scalar ( $c$ ) is multiplied with one row of a matrix  $A$ , then the determinant of the new matrix will be  $c$  times the determinant of  $A$ .]

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 8

1 point

1 point

If  $A = \begin{bmatrix} 2 & -11 & 1 \\ 1 & -3 & 4 \\ -1 & 0 & 2 \end{bmatrix}$  and  $B = \begin{bmatrix} 13 & -2 & 0 \\ 0 & 1 & -1 \\ 3 & 4 & 2 \end{bmatrix}$ , then choose the set of

correct options.

☐

$\det(A) = -9$  and  $\det(B) = -3$

☐

$\det(A) = 9$  and  $\det(B) = 3$

☐

$\det(A) = -9$  and  $\det(B) = 3$

☐

$\det(AB) = -27$  and  $\det(BA) = 27$

☐

$\det(AB) = \det(BA) = -27$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\det(A) = -9$  and  $\det(B) = 3$

$\det(AB) = \det(BA) = -27$

1 point

Let  $A$  be a  $2 \times 2$  matrix, which is given as  $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ . Define the following

matrices :

$B = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$ ,  $C = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$

$D = \begin{bmatrix} a_{11} + a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$ ,  $E = \begin{bmatrix} a_{11} - a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{bmatrix}$



Which of the matrices among  $B, C, D$ , and  $E$  have the same determinant as that of the matrix  $A$ , for any real numbers  $a_{11}, a_{12}, a_{21}, a_{22}$ ?

☐

$BB$  and  $DD$

☐

$BB$  and  $EE$

☐

$BB$  and  $CC$

☐

$DD$  and  $EE$

☐

$CC$  and  $EE$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$BB$  and  $CC$

Let  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ r a_{31} & r a_{32} & r a_{33} \end{bmatrix}$  be a matrix and  $r, s, t \neq 0$ . Find  $\det(A)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

Suppose  $A = \begin{bmatrix} -14 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 & -1 \\ 4 & \alpha & 0 \\ 0 & -3 & -5 \end{bmatrix}$ . If

$\det(AB) = 32$ , then find the value of  $\alpha$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

#### 4.2 Level 2:

**(Use this information to answer questions 6 and 7)**

Let  $A(x_1, y_1)$ ,  $B(x_2, y_2)$  and  $C(x_3, y_3)$  be the vertices of a

triangle. The area of the triangle  $ABC$  is given by  $\frac{1}{2} |\det(D)|$  square units, where

$$D = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

If  $P(-1, 1)$ ,  $Q(1, 1)$  and  $R(1, 0)$  are the vertices of a triangle, then find the area (in square units) of the triangle  $PQR$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Let  $P(x-1, 2x)$ ,  $Q(0, x-2)$  and  $R(-x+1, 1)$  be the vertices of a triangle. If the area of the triangle is 55 square units and  $x > 0$ , then choose the set of correct options

☐

length of side  $PQ = \sqrt{29}$

☐

units.

☐

length of side  $PR = \sqrt{40}$

☐

units.

☐

length of side  $QR = 2$

☐

length of side  $QR = 1$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

length of side  $PQ = \sqrt{29}$

☐

units.

length of side  $QR = 2$

1 point

Find out the correct value of  $\det(A)$ , where  $A = \begin{bmatrix} 120102020 \times 2030 \\ 120202030 \times 2010 \\ 120302010 \times 2020 \end{bmatrix}$

☐

[Hint: Observe that the given matrix is of the form:  $\begin{bmatrix} 1abc \\ 1bca \\ 1cab \end{bmatrix}$ ]

☐

00

☐

200200

☐

-2000-2000

☐

20002000

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

20002000

1 point

Let  $AA$  be a square matrix of order 33 and  $BB$  be a matrix that is obtained by adding 22 times the first row of  $AA$  to the third row of  $AA$  and adding 33 times the second row of  $AA$  to the first row of  $AA$ . What is the value of  $\det(6A^2 B^{-1})\det(6A2B-1)$ ?

☐ $6 \det(A)6 \det(A)$ ☐ $6 \det(A)\det(B)6 \det(A)\det(B)$ ☐ $6^3 \det(A)63 \det(A)$ ☐ $6^3 \det(A)\det(B)63 \det(A)\det(B)$ **No, the answer is incorrect.****Score: 0****Accepted Answers:** $6^3 \det(A)63 \det(A)$ 

Suppose  $A = \begin{bmatrix} 192021 \\ 202119 \\ 211920 \end{bmatrix}$ . Find the value of  $\frac{1}{60} \det(A^2)601 \det(A2)$ .

[[ You can try to compute the determinant of a matrix of the form:  $\begin{bmatrix} abc \\ bca \\ cab \end{bmatrix}$  ]]

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 540

1 point